



DATA ANALYTICS

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Unit 2: Regression Analysis

Lecture 20-21: Simple Linear Regression

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Data Analytics

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Lectures 20-21: Simple Linear Regression

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- Introduction to Regression
 - What is Regression?
- Types of Regression
- Simple Linear Regression (SLR)
- SLR Model Building
- Ordinary Least Squares Assumptions
- Interpretation of coefficients

- Regression is a tool for finding **existence of association relationship** between dependent variable Y and one or more independent variables $X_1, X_2, \dots, X_n$
- Relationship can be **linear** or **non-linear**
- **Dependent variable**: measures outcome of a study; responds to change in independent variables (outcome variable, response variable)
- **Independent variable**: explains the change in a response variable (explanatory variable)
- Regression: set values of explanatory variable and see how it affects the response variable

- Regression is a supervised learning algorithm under machine learning terminology
 - Requires knowledge of both Y and X in training set
- Regression is a statistical technique to determine if there is a relationship between a dependent variable (Y) and a set of independent variables ($X_1, X_2, \dots, X_n$)
- Regression (association) does not imply causation, even though terms like “independent” and “dependent” are used
- Regression is a tool that lets us predict the value of the dependent variable given the values of the independent variable
- Regression used for generating new hypotheses and validating the hypotheses

Dependent Variable	Independent Variable
Explained Variable	Explanatory variable
Regressand	Regressor
Predictand	Predictor
Endogenous Variable	Exogenous Variable
Controlled Variable	Control Variable
Target Variable	Stimulus Variable
Response Variable	Feature
Outcome Variable	

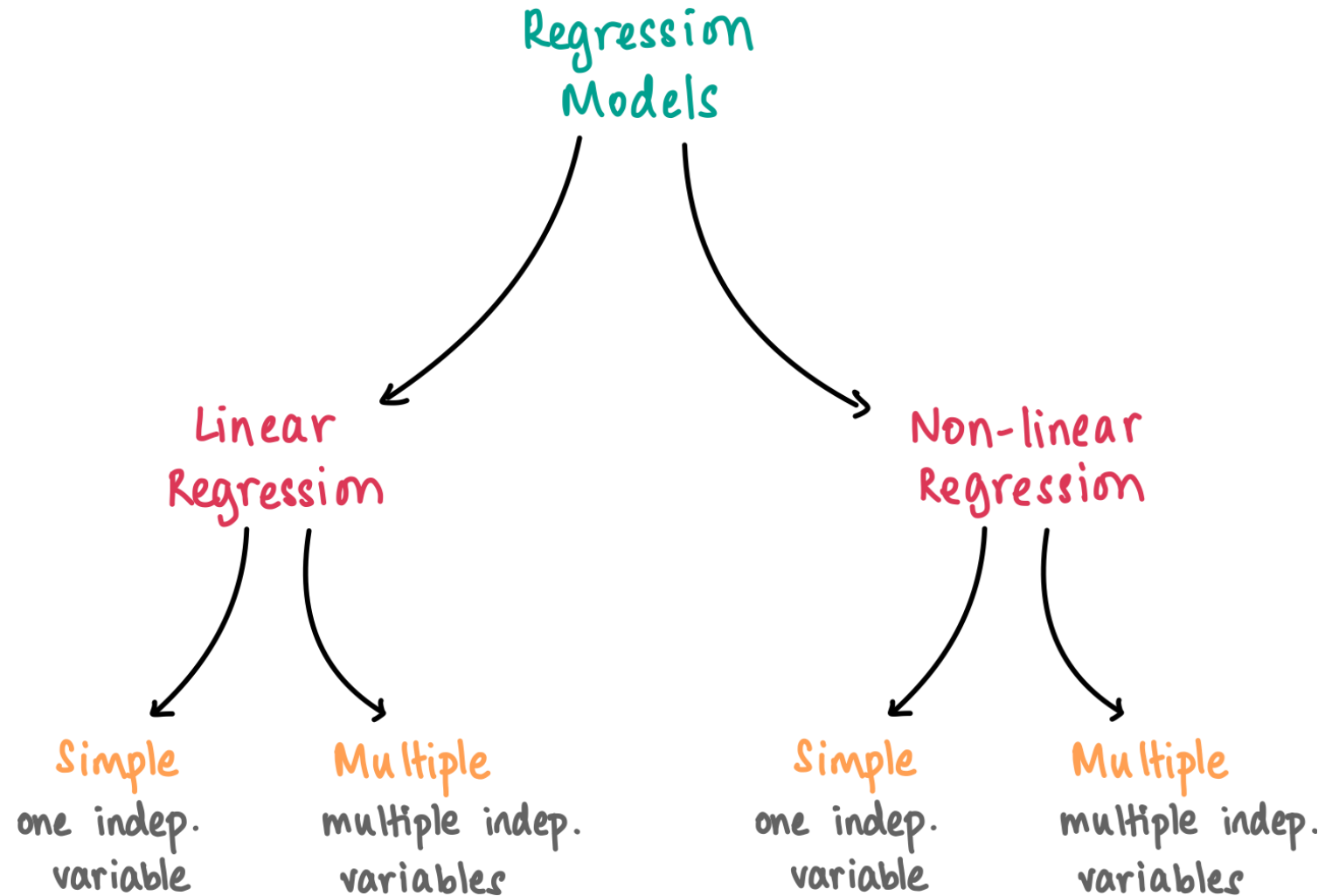
- A common business use case of regression is for predicting prices of goods and assets. Say you are trying to predict the prices of houses in a certain neighborhood of a certain suburb. Think of possible features/factors that may be used as explanatory variables.

- Some features maybe

- Square footage
- Average income
- ...



Source: Getty Images



- **Simple Linear Regression:** single independent variable, functional relationship between the dependent variable and the regression coefficient is linear

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

- **Multiple Linear Regression:** multiple independent variables, functional relationship between the dependent variable and the regression coefficients is linear

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

- **Non-linear Regression:** non-linear relationship between dependent variable and regression coefficients

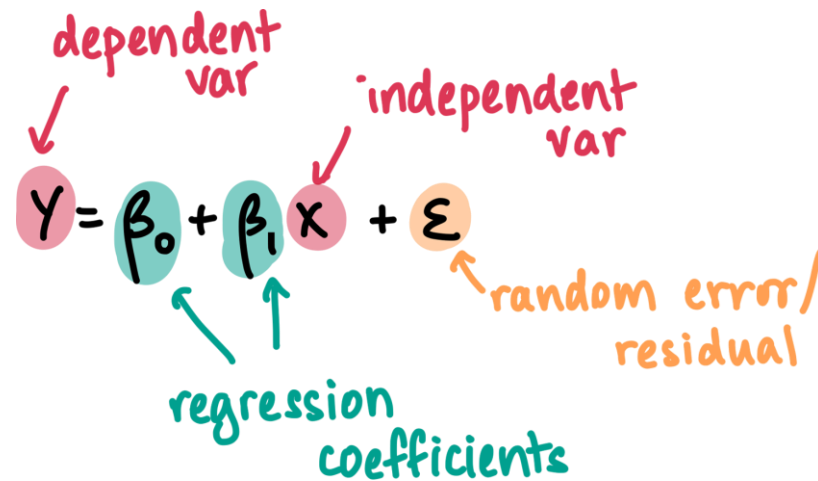
$$Y = \beta_0 + \frac{1}{\beta_1 + \beta_2 X_1} + X_2^{\beta_3} + \epsilon$$

- Linear relationship between dependent variable and regression coefficients (β_0 and β_1)
- Not defined as a linear relationship between dependent variables and independent variables
- The below regression is also a linear regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 X_1 + \beta_3 X_1^2 + \epsilon$$



- Functional form of SLR is $Y = \beta_0 + \beta_1 X + \epsilon$



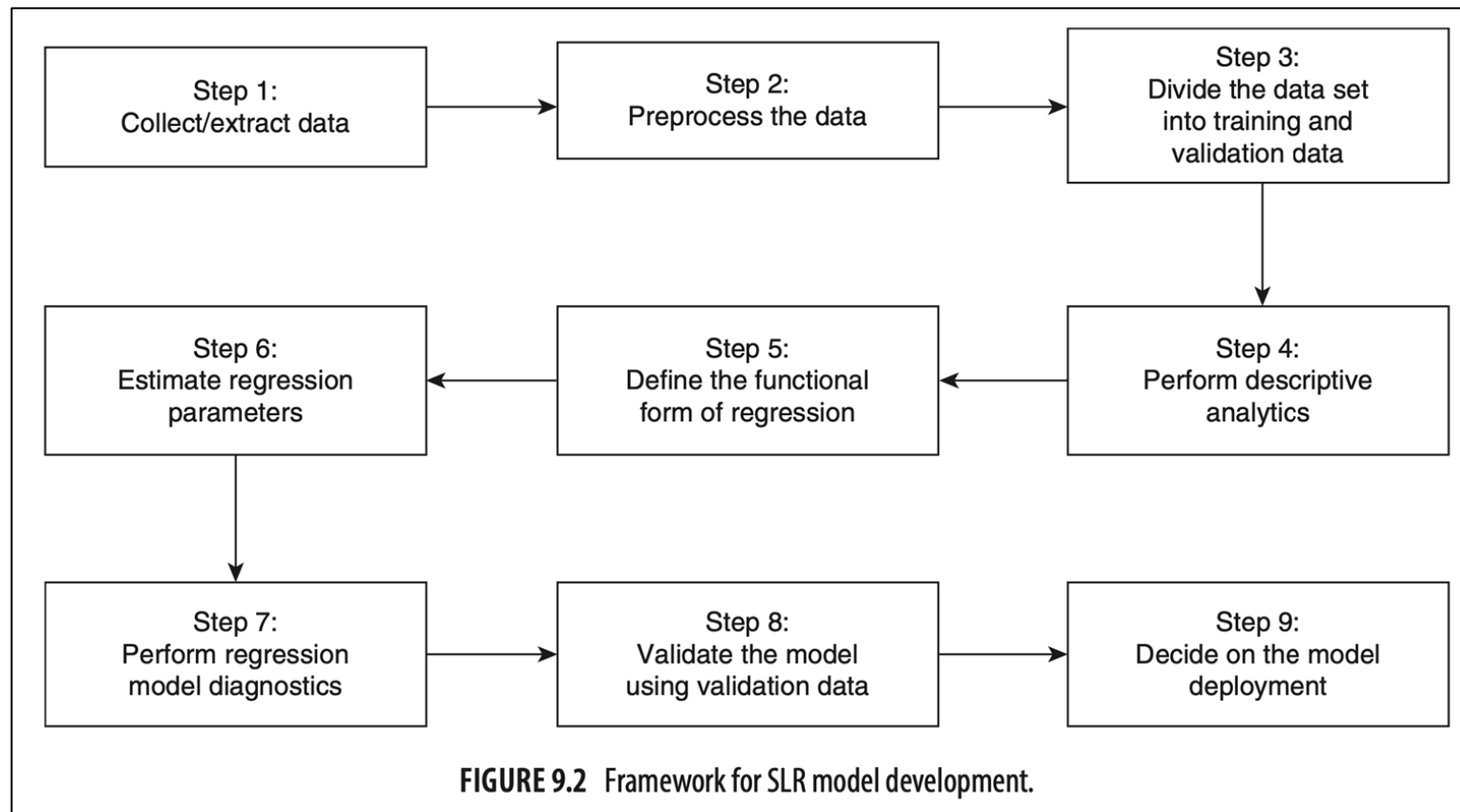
The diagram shows the equation $Y = \beta_0 + \beta_1 X + \epsilon$ with handwritten annotations in various colors:

- Y is labeled "dependent var" with a red arrow.
- X is labeled "independent var" with a red arrow.
- β_0 and β_1 are labeled "regression coefficients" with green arrows.
- ϵ is labeled "random error/residual" with an orange arrow.

SLR attempts to explain the changes in the value of response variable Y using the knowledge of the values of explanatory variables X . Thus, the equation $b_0 + b_1 X_i$ gives the predicted value of Y_i for a given value of X_i , whereas the term e_i is the error in predicting the values of Y_i . In fact $b_0 + b_1 X_i$ is the conditional expected value of Y_i for a given value of X_i .

- Regression is a statistical relationship (not mathematical) and is, hence, not exact (calculated as a best-fit function)
- Since it is inexact, there is an error term called the **residual** or **random error**
- Such errors cannot be explained by the model

● 9-step process



● Step 1: Collect data

- The first step in building a regression model is to collect and/or extract data from different sources for the identified problem (or KPI – Key Performance Indicators)
- Can be time consuming and expensive

● Step 2: Preprocess data

- Pre-processing the data is an important stage in the regression model building
- Before the model is built, it is also essential to ensure the quality of the data for issues such as reliability, completeness, usefulness, accuracy, missing data, and outliers.
- Dummy variables for nominal data

● Step 3: Divide into Train and Validation

- While choosing best model, select based on performance in validation set

● Step 4: Perform descriptive analysis

- It is always a good practice to perform descriptive analytics before moving to predictive analytics model building
- Visualization techniques to see distribution of data

- Step 5: Define the functional form of relationship

- Scatter plots

- Step 6: Estimate regression parameters

- The method of Ordinary Least Squares (OLS) is used to estimate the regression parameters

- Step 7: Perform regression model diagnostics

- Before it can be deployed it is necessary that the regression model created is validated for all model assumptions including the definition of the functional form

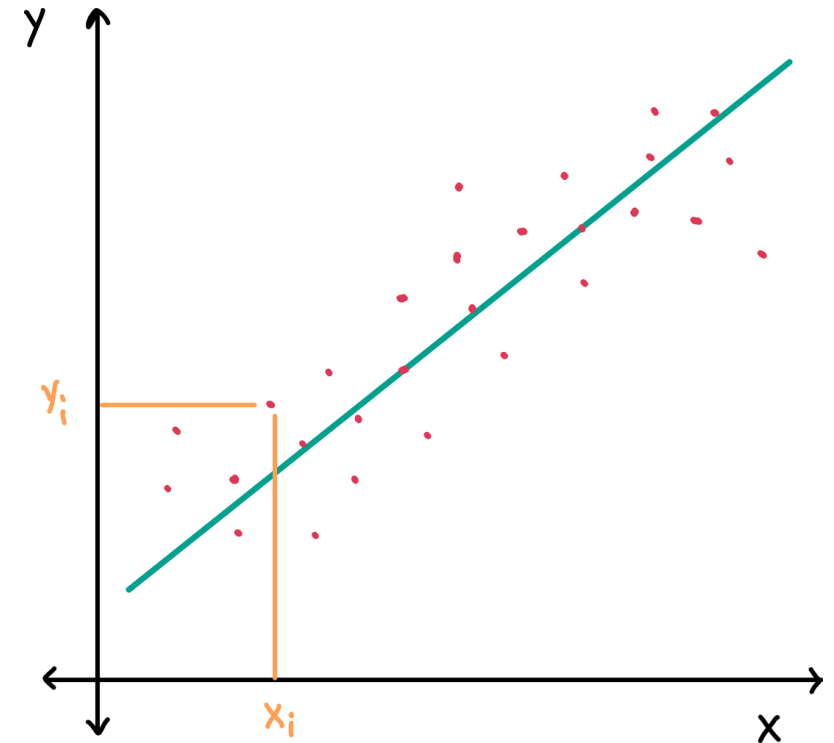
- Step 8: Validate model using validation set

- Model may be cross-validated using multiple training and test data sets
- Check for overfitting

- Step 9: Decide on model deployment

- Estimation of regression coefficients involves solving a system of linear equations
- Best equation to represent relationship between observations
- Provides the **Best Linear Unbiased Estimate** (BLUE) – $E[\beta - \hat{\beta}] = 0$
- OLS: minimize sum of squared errors (SSE) over all n observations

$$\text{SSE} = \sum_{i=1}^n \epsilon_i^2$$
$$\text{SSE} = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$



1. The regression model is **linear in regression parameters**
1. The **explanatory variable**, X , is assumed to be **non-stochastic** (i.e., X is deterministic)
1. The **conditional expected value of the residuals**, $E(e_i | X_i)$, is **zero**
1. In case of **time series** data, **residuals are uncorrelated**, that is, $\text{Cov}(e_i, e_j) = 0$ for all $i \neq j$
1. The **residuals**, e_i , follow a **normal distribution**
1. The **variance of the residuals**, $\text{Var}(e_i | X_i)$, is **constant** for all values of X_i . When the variance of the residuals is constant for different values of X_i , it is called **homoscedasticity**. A non-constant variance of residuals is called **heteroscedasticity**.

- Equate partial derivatives of SSE wrt regression coefficients to 0 (can skip derivation; just for understanding)

- For β_0

$$\frac{\partial \text{SSE}}{\partial \beta_0} = \sum_{i=1}^n -2(Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\underbrace{\sum_{i=1}^n Y_i}_{n\bar{Y}} - n\beta_0 - \beta_1 \underbrace{\sum_{i=1}^n X_i}_{n\bar{X}} = 0$$

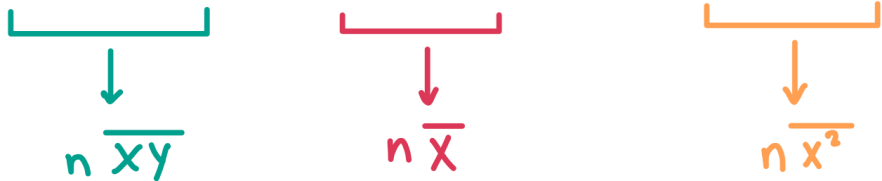
$$\cancel{n}\bar{Y} - \cancel{n}\beta_0 - \cancel{n}\beta_1\bar{X} = 0$$

$$\bar{Y} - \beta_0 - \beta_1\bar{X} = 0 \longrightarrow (1)$$

● For β_1

$$\frac{\partial \text{SSE}}{\partial \beta_1} = \sum_{i=1}^n -2X_i(Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$$



$$n \overline{xy} - n \beta_0 \bar{x} - n \beta_1 \bar{x}^2 = 0$$

$$\overline{xy} - \beta_0 \bar{x} - \beta_1 \bar{x}^2 = 0 \longrightarrow (2)$$

● Solving (1) and (2)

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} \quad \text{from (1)}$$

$$\beta_0 = \frac{\overline{XY} - \beta_1 \overline{X^2}}{\bar{X}} \quad \text{from (2)}$$

$$\bar{X} \bar{Y} - \beta_1 (\bar{X})^2 = \overline{XY} - \beta_1 \overline{X^2}$$

$$\beta_1 (\overline{X^2} - (\bar{X})^2) = \overline{XY} - \bar{X} \bar{Y}$$

$$\hat{\beta}_1 = \frac{\overline{XY} - \bar{X} \bar{Y}}{\overline{X^2} - (\bar{X})^2}$$

estimate for β_1

$$\hat{\beta}_0 = \bar{Y} - \beta_1 \bar{X}$$

estimate for β_0

● Rewriting

$$\hat{\beta}_1 = \frac{n(\overline{XY} - \bar{X} \bar{Y})}{n(\overline{X^2} - (\bar{X})^2)} = \frac{\sum_{i=1}^n x_i y_i - \bar{X} \bar{Y}}{\sum_{i=1}^n x_i^2 - \bar{X}^2}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - x_i \bar{Y}}{\sum_{i=1}^n x_i^2 - x_i \bar{X}} \rightarrow (3)$$

(why are these equivalent?)

$$\sum_{i=1}^n \bar{X} \bar{Y} = \sum_{i=1}^n x_i \bar{Y} = \sum_{i=1}^n \bar{X} y_i$$

$\Rightarrow$ difference of any 2 terms = 0

- Adding 0 to the numerator and denominator of (3)

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - x_i \bar{y} + \bar{x} \bar{y} - \bar{x} y_i}{\sum_{i=1}^n x_i^2 - x_i \bar{x} + (\bar{x})^2 - \bar{x} x_i}$$

0 (see prev page) ↙

↗ 0

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

- In terms of Pearson's Coefficient

Pearson Correlation Coefficient

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\hat{\beta}_1 = r \frac{\sigma_y}{\sigma_x}$$

Example

S. No.	Percentage in Grade 10	Salary	S. No.	Percentage in Grade 10	Salary
1	62	270000	26	64.6	250000
2	76.33	200000	27	50	180000
3	72	240000	28	74	218000
4	60	250000	29	58	360000
5	61	180000	30	67	150000
6	55	300000	31	75	250000
7	70	260000	32	60	200000
8	68	235000	33	55	300000
9	82.8	425000	34	78	330000
10	59	240000	35	50.08	265000
11	58	250000	36	56	340000
12	60	180000	37	68	177600
13	66	428000	38	52	236000
14	83	450000	39	54	265000
15	68	300000	40	52	200000
16	37.33	240000	41	76	393000
17	79	252000	42	64.8	360000
18	68.4	280000	43	74.4	300000
19	70	231000	44	74.5	250000
20	59	224000	45	73.5	360000
21	63	120000	46	57.58	180000
22	50	260000	47	68	180000
23	69	300000	48	69	270000
24	52	120000	49	66	240000
25	49	120000	50	60.8	300000

Table provides the salary of 50 graduating MBA students of a Business School in 2016 and their corresponding percentage marks in grade 10. Develop a linear regression model by estimating the model parameters.

- Tip: if you have a Casio fx-991EX or 991ES, you could use the Statistics menu (number 6), $y=a+bx$ and type out all the x and y values

let x = class 10 percent
 y = salary

$$\hat{\beta}_0 = 61555.3553$$

$$\hat{\beta}_1 = 3076.1774$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{y} = 61555.3553 + 3076.1774 x$$

- Important for understanding the impact of change in the values of explanatory variables on the response variable
- The interpretation will depend on the **functional form** of the relationship between the response and the explanatory variables

1. Interpretation in $Y = \beta_0 + \beta_1 X$

- **B1** is the change in Y for a unit change in X
- **B0** is the expected value of Y when X = 0

$$\beta_1 = \frac{\partial Y}{\partial X}$$

$$\beta_0 = E(Y | X=0)$$

1. Interpretation in $Y = \beta_0 + \beta_1 \ln(X)$

- **B1** is the change in Y for percentage change in X
- **B0** is the expected value of Y when $\ln(X) = 0$ or X = 1

$$\frac{\partial Y}{\partial X} = \frac{1}{X} \beta_1 \Rightarrow \beta_1 = \frac{\partial Y}{(\partial X/X)}$$

$$\beta_0 = E(Y | \ln(X) = 0)$$

9.5.3 | Interpretation of β_0 and β_1 in $\ln(Y) = \beta_0 + \beta_1 X$

$$\beta_0 = E(\ln(Y)|X=0)$$

Differentiating the equation with respect to X , we get

$$\frac{1}{Y} \frac{\partial Y}{\partial X} = \beta_1 \Rightarrow \frac{\partial Y/Y}{\partial X} = \beta_1$$

β_1 is the percentage change in Y for a unit change in the X value. An alternative interpretation can be derived as follows:

$$Y = e^{(\beta_0 + \beta_1 X)}$$

Change in Y for one unit change in X is given by

$$e^{(\beta_0 + \beta_1 (X+1))} - e^{(\beta_0 + \beta_1 X)} = e^{(\beta_0 + \beta_1 X)} \times (e^{\beta_1} - 1) \quad (9.16)$$

That is, one unit change in X changes Y by a factor of $e^{(\beta_0 + \beta_1 X)} \times (e^{\beta_1} - 1)$.

9.5.4 | Interpretation of β_0 and β_1 in $\ln(Y) = \beta_0 + \beta_1 \ln(X)$

Differentiating the equation with respect to X , we get

$$\frac{1}{Y} \frac{\partial Y}{\partial X} = \frac{\beta_1}{X} \Rightarrow \beta_1 = \frac{\partial Y/Y}{\partial X/X}$$

β_1 is the percentage change in Y for a percentage change in X .

● Which of the following are linear regression models?

(a) $Y = \beta_0 + \beta_1 X^2$

(b) $Y = \beta_0 + [1/(1+\beta_1)] X$

(c) $Y = \beta_0 + \beta_1 X$

(d) $\ln(Y) = \beta_0 + \beta_1 \ln(X)$

● Solution

~~(a)~~ $Y = \beta_0 + \beta_1 X^2$

(b) $Y = \beta_0 + [1/(1+\beta_1)] X$

~~(c)~~ $Y = \beta_0 + \beta_1 X$

~~(d)~~ $\ln(Y) = \beta_0 + \beta_1 \ln(X)$

- What is the meaning of “Heteroscedasticity”?
 1. The variance of errors is not constant
 2. The variance of the dependent variable is not constant
 3. The errors are not linearly independent of one another
 4. The errors have nonzero mean
- Solution
- **The variance of the errors is not constant**

- State True or false:

In an OLS set-up, if the errors follow normal distribution, then Y will also follow a normal distribution.

- Solution:
- True, since B_0 and B_1 are parameters and X is deterministic.

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022 [Chapter 9](#)
- U. Dinesh Kumar's slides
- Getty Images



THANK YOU

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